THE UNIVERSITY OF HONG KONG
Faculty of Business & Economics
2019-2020 2nd Semester Examination

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Finance: FINA2322
Derivatives
Sub-class E, F, G: Dr Yang Liu

April 14, 2020 8:00-9:30 p.m.

There are two parts in the exam.

Part I. 7 Multiple Choice Questions (5 point each, 35 points in total)

Part II. 7 Short Questions (65 points in total)

The total points are 100.

Total Pages of the Exam Paper (including this cover page): 9

- **You have to write down your Name and UID on the Answer Script.**

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1. All the work will be my own, and I will not plagiarize from anyone else;

2. I will not obtain or seek to obtain an unfair advantage by communicating or attempting to communicate with any other person during the examination; neither will I give or attempt to give assistance to another student in taking the examination;

B. I understand that students who are suspected of violating this pledge are liable to be referred to the Disciplinary Committee, and maybe subject to disciplinary action such as suspension of studies or expulsion from the University.

Case 1 Hong Kong Offshore Oil Corporation (HKOOC)

After the huge virus shock and the global negotiation between major oil producing countries, the oil price has fell from $60 in January to $20 in March. You are a bank officer and a client, Hong Kong Offshore Oil Corporation (HKOOC) has come to you for advice in risk management.

First of all, HKOOC sells oil in operation. They are considering whether or not to hedge their oil price risk. You explained to them what the advantages of hedging are and also explained reasons not to hedge.

MC Question 1 (5 points)

Which of the following is a reason that HKOOC wants to hedge risk?

A) Monitoring costs
B) Transaction costs
C) Insurance costs
D) External financing costs

MC Question 2 (5 points)

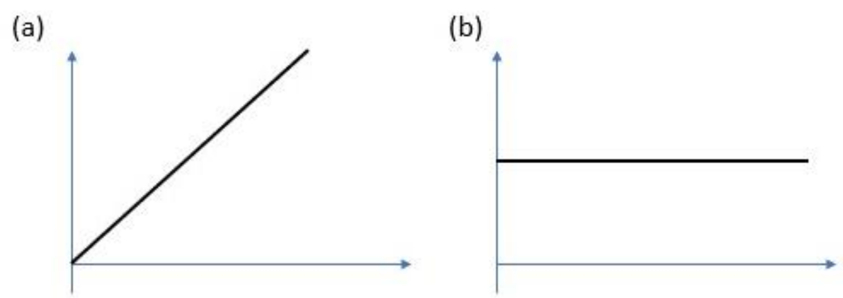
HKOOC has an inherent ___(1)___. They could hedge his position by ___(2)___.

	(1)	(2)
A)	Short position in the oil price.	Long a put on oil.
B)	Long position in the oil price.	Buy a collar on oil.
C)	Short position in the oil price.	Long oil forward.
D)	Long position in the oil price.	Short a put on oil.

Short Questions 1 (6 points)

HKOOC has scheduled to sell oil exactly 6 months from now (1 April). HKOCC is considering a strategy that would hedge all the oil price risk (strategy 1) and generate a constant revenue. The effective 6-month interest rate is 3.5%.
The oil price for today (April 1) is $20 per barrel.

(a) What is the unhedged revenue per barrel? Draw the payoff diagram. (3 points)
(b) What is the hedged revenue per barrel? Draw the payoff diagram. (3 points)



After examining the payoff of strategy 1, HKOOC found that the strategy limits too much revenue of the company. You explained some methods to preserve the upside potential.

MC Question 3 (5 points)

Which of the following can preserve the upside potential and protect the downside risk at the same time?

A) Long Collar
B) Long Put
C) Bear Spread
D) Cap

MC Question 4 (5 points)

HKOOC decides to use a forward contract to hedge his position. Suppose 3 months later, the spot oil price is 20.4, and the effective 3-month interest rate is 1.8%. What will be the value of HKOOC's forward position then?

A) Zero
B) Positive
C) Negative
D) Unknown

Your client is excited to learn about the forward contract. He asked about whether he can use the forward contract to predict the oil price in the future.

Short Question 2 (4 points)

Explain whether the forward price can be used to predict the oil price.

No.
Forward price undervalues the expected oil price as the forward price is calculated based on risk-free interest rate instead of expected rate of return (which is used to calculate the expected stock price). Since expected return is higher than risk free rate, the forward price is smaller than the expected oil price and hence forward is a biased predictor of the expected oil price.

HKOOC is also considering entering futures contracts and asks you to explain the differences between futures and forward.

<u>MC Question 5 (5 points)</u>

Regarding the difference between forward and futures contracts, which of the following statements is correct?

A) Futures are exchange-traded forward contracts
B) Futures contracts can be customized
C) Futures are marked to market every week
D) Futures price is always the same as forward price

<u>MC Question 6 (5 points)</u>

You have some thoughts when you explain to HKOOC about forwards and futures.

1. Futures contract has more default risk than a forward contract because the former has a higher contract size.
2. When you take a long position in the forward contract, the initial value of the contract is zero.
3. The annualized forward premium is higher when the dividend yield of the underlying asset is higher.
4. When you buy a stock index forward, you expect to earn compensation for the time value of money.
5. When you buy a stock index forward, you expect to earn compensation for risk.
6. When there are transaction costs, the no arbitrage forward price becomes a range instead of a number.

How many of the above statements are true?

A) 1
B) 2
C) 3
D) 4

Short Questions 3 (13 points)

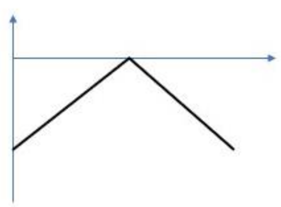
HKOOC has a trading department to speculate on oil price. The department expects oil price to be very stable in the coming 6 months, i.e. stays at the price of \$20/bbl.

Eventually, HKOOC decides to profit from the expected situation of oil price and is considering using straddle to help capture the profit (strategy 2).

The following options mature in 6 months and the effective 6-month interest rate is 3.5%. The oil price for today (April 1) is \$20 per barrel. Below is the relevant call and put premiums:

Strike	Call premium	Put Premium
18	3.62	1.01
20	2.34	1.66
22	1.42	2.68

(a) Draw the payoff diagram of strategy 2. (2 points)



(b) Find out the cost of strategy 2. (2 points)
-2.34 -1.66 = -4.00

(c) From the payoff diagram, what is the potential risk of using strategy 2? (2 points)
There could be huge losses under extreme price movement.

(d) As the counterparty of HKOOC, you faces some risks and want to hedge them using 18-22 strangle. Find the cost of strangle. (3 points)
-1.01-1.42 = -2.43

(e) What is the range of oil prices such that the profit of strangle is positive? (4 points)
FV of cost = 2.43(1+3.5%) = 2.51505
Lower bound: 18-2.51505 = 15.48495
Upper bound: 22+2.51505 = 24.51505

Short Questions 4 (14 points)

HKOOC wants to borrow $500,000 9 months later for 6 months with a Forward Rate Agreement. The following table shows bond market information.

Maturity (month)	Zero Coupon Bond Price
3	0.966
6	0.951
9	0.933
12	0.911
15	0.884

(a) What is the forward rate of the FRA (effectively for 6 months)? (4 points)
0.933/0.884-1 = 5.5430%

(b) Suppose 9 months later, the 6-month annualized spot rate is 7%. What is the settlement amount of the FRA if settled at initiation? (4 points)
((3.5%-5.5430%)*500000)/(1+3.5%) = -9869.4996

(c) If in the market, the above forward contract effective 6-month FRA rate is 5.5%. Can you take an arbitrage profit from it? Assume the size of contract is $500,000. (Use the following table to show the cashflows.) (You can assume the FRA is settled at initiation or in arrears) (6 points)

Position	T = 0	T = t	T = T
Long FRA	0	0	(r-5.5%)*500000
Borrow at time = t	0	500000	-(1+r)*500000
Borrow at t=0, repay at t=9 month	+500000*0.933 = 466500	-500000	0
Lend at t=0, receive at t=15month	-466500	0	+466500/0.884 =527714.93
Total	0	0	214.93

Case 2 Derivatives Asset Management

You are a fund manager in Derivatives Asset Management Fund (2322.HKU). You manage a portfolio consists of stocks, bonds and currencies. The following table shows information of Treasury bond market in Hong Kong (denominated in HKD) and Australia (denominated in AUD).

Year	Continuously Compounded Zero-yield (HKD)	One-year implied forward rate (HKD)	Par coupon rate (HKD)	Zero-coupon bond yield (AUD)
1		2%		4%
2			3%	5%
3	4%			6%

Short Question 5 (6 points)

(a) What is the price of a one-year zero-coupon bond issued one year later (i.e. at t=1)? Assume the FV of this bond is 1000. (3 points)

$$\frac{3}{1.02} + \frac{103}{1.02(1 + r_0(1,2))} = 100$$

$$r_0(1,2) = 4.0404\%$$

$$Bond\ price = P_0(1,2) = \frac{1000}{1 + 4.0404\%} = 961.1650$$

(b) What is the PVBP of the bond in (a)? (3 points)

$$PVBP = \frac{DUR_{MAC}}{1 + y}(price)(\Delta y) = \frac{1}{1.040404}(961.1650)(0.01\%) = 0.09238$$

Short Question 6 (11 points)

Your bond portfolio has a large position in a 3-year corporate bond issued by a real estate company Yellow River Holdings. The bond pays coupon annually with a face value of \$100 million. The current bond portfolio value is \$100 million with a yield to maturity of 8%. You are worried about the interest rate hikes that might affect the Yellow River bond.

(a) What is the coupon rate of the bond? (2 points)
8%, since the bond is priced at par, the coupon rate is equal to the yield to maturity.

(b) What is the modified duration of the bond? (2 points)

$$DUR_{MAC} = \frac{\frac{8}{1.08} + \frac{8(2)}{1.08^2} + \frac{108(3)}{1.08^3}}{100} = 2.7833$$

$$DUR_{mod} = \frac{2.7833}{1.08} = 2.5771$$

(c) If the yield of Yellow River bond increases by 1 basis point, what is the change in the bond's value approximated by duration? (2 points)

$$change = -DUR_{mod}(price)(\Delta y) = -2.5771(100m)(0.01\%) = -25770.9699$$

(d) Assume that the yield of Yellow River bond and the yield of the 3-year zero-coupon Treasury bond always fluctuate by the same amount. Assume the FV of the 3-year zero-coupon Treasury bond is 1000. Based on the information above, how many 3-year zero-coupon bonds should you long or short to hedge against the interest rate risk of the Yellow River bond? (5 points)

$$PVBP_{Treasury} = \frac{DUR_{MAC}}{1+y}(price)(\Delta y) = \frac{3}{e^{0.04}}\left(\frac{1000}{e^{0.04(3)}}\right)(0.01\%) = 0.2556$$

$$Unit\ of\ Treasury\ bond = -\frac{25770.9699}{0.2556} = -100808.38$$

Answer: Short 100808.38 units of bond.

MC Question 7 (5 points)

You have some thoughts when you develop the investment strategies.

1. As a lender, you can hedge the interest rate risk by taking a short position in Eurodollar futures.
2. A coupon bond is more sensitive to interest rate risk than a zero-coupon bond with same time to maturity.
3. The duration of the bond can only provide an approximation on the change in bond price due to interest rate. It is because duration also changes with the interest rate.
4. A repurchase agreement allows the investors to synthetically create a borrowing position.

How many of the above statements are true?

A) 1
B) 2
C) 3
D) 4

Short Question 7 (11 points)

(a) You also trade on the currency market. The current spot exchange rate is 4.7 HKD/AUD. The strategy of borrowing a low-rate currency and lending a high-rate currency is called a carry trade. Show how to conduct a carry trade for 1 year. Start with HKD 1000 at t=0. Denote x_T the exchange rate a year from now.(6 points)

Positions	T = 0		T = 1	
	HKD	AUD	HKD	AUD
Borrow HKD at t=0 for 1 year	1000	0	-1000(1.02) = -1020	0
Convert HKD to AUD	-1000	+1000/4.7 = 212.77	0	0
Lend AUD at t=0 for 1 year	0	-212.77	0	+212.77(1.04) = 221.28
Convert AUD to HKD at t=1			$+221.28x_T$	-221.28
Total	0	0	$221.28x_T$ - 1020	0

(b) Exchange rates are believed to follow a "random walk", which means the expected exchange rate is the same as the current exchange rate. Random walk is a statistical term vividly capturing the idea that something walks randomly so the expected value of the future exchange rate is the current exchange rate. Based on the answer in (a), how much (in HKD) do you expect to get back at t=1? What is the risk? (3 points)

$$\textit{expected amount} = 221.28\mathrm{E}(x_T) - 1020 = 221.28(4.7) - 1020 = 20$$

The risk is exchange rate risk. Carry trade is a bet on the high rate currency (AUD) will not depreciate much.

(c) To mitigate the risk, you can lock in the exchange rate next year using a currency forward. What is the average return of a hedged (covered) position? (2 points)

$$F_{0,1} = \frac{4.7(1.02)}{1.04} = 4.6096$$

$$\textit{return} = 221.28(4.6096) - 1020 = 0$$

OR

With a forward contract, the carry trade position does not have any risk (as currency forward hedges away the exchange rate risk) and does not have any initial investment. Hence, both time value of money and risk will not be compensated and the return would be zero.